

A MAXIMUM LIKELIHOOD ESTIMATION PROCEDURE WHEN MODELLING CATEGORICAL DATA IN TERMS OF CROSS-PRODUCT RATIOS

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Summary: Data cross-classified in a contingency table can be modelled in terms of cross-product ratios. Well-known models such as the loglinear model, the models of quasi-symmetry and quasi-independence can be expressed in terms of cross-product ratios. This paper provides a maximum likelihood estimation procedure for estimating the expected frequencies subject to the constraints imposed on the frequencies through the cross-product ratios.

1. Introduction

There has in recent years, been a significant development in the field of analyzing categorical data. Many models have been proposed and the class of models called generalized linear models, introduced by Nelder and Wedderburn (1972), has also had a large contribution to the variety of models available for modelling data. When models do not have closed form expressions for the maximum likelihood (ML) estimates, then iterative algorithms are required

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for solving the likelihood equations, thus yielding the parameter estimates and fitted values. One of the first methods used for finding the ML estimates was the iterative proportional fitting (IPF) algorithm, which was originally introduced by Deming and Stephan (1940). Haberman (1974, 1978, 1979) also makes extensive use of IPF in his work on loglinear models. More recently the Newton Raphson procedure has been used for finding the ML estimates of the parameters in a model. Agresti (1990) discusses the advantages and disadvantages of the above-mentioned procedures.

An alternative ML estimation procedure proposed by Matthews and Crowther (1995), provides the ML estimate for the mean vector of the multivariate exponential family when modelling in terms of certain constraints on the elements of the mean vector. This procedure provides a very flexible method for finding the ML estimates for the mean values in a model provided that the covariance matrix of the variables can be computed.

This method is particularly useful when a set of natural constraints is placed on the cell frequencies of a contingency table. There are a number of models in which constraints are placed on the cross-product ratios. The estimation procedure which we have proposed, makes direct use of these constraints without having to construct a parametric form of the model, which is the traditional solution followed to implement the Newton Raphson procedure, thus obtaining the ML estimates. The emphasis of this paper is how the natural constraints on the cell frequencies may be used to find the ML estimates of the expected frequencies instead of using the parametric forms of the models implied by the constraints. The loglinear model as well as the quasi-

independence and quasi-symmetry models are easily and naturally expressed in terms of constraints on the cross-product ratios and will serve as examples.

Firstly it will be beneficial to present some notation and the ML estimation procedure of Matthews and Crowther (1995), as it is the foundation of the work that follows. Details of the proof and other applications may be found in the latter reference.

2. A maximum likelihood estimation procedure

Let $\mathbf{X}$ be a $p \times 1$ random vector and $\boldsymbol{\theta}$ be a $p \times 1$ vector of parameters. The probability function for the exponential family is given by

$$\begin{aligned} p(\mathbf{x}, \boldsymbol{\theta}) &= a(\boldsymbol{\theta})b(\mathbf{x}) \exp\{\mathbf{x}'\boldsymbol{\theta}\}, \mathbf{x} \in \mathbb{R}^p, \boldsymbol{\theta} \in \mathcal{N} \\ &= b(\mathbf{x}) \exp\{\mathbf{x}'\boldsymbol{\theta} - \kappa(\boldsymbol{\theta})\}, \end{aligned} \quad (2.1)$$

where $\kappa(\boldsymbol{\theta}) = -\ln a(\boldsymbol{\theta})$ and $\mathcal{N}$ is the natural parameter space for the canonical parameter $\boldsymbol{\theta}$. The mean vector and covariance matrix of $\mathbf{X}$ are given by

$$E(\mathbf{X}) = \boldsymbol{\mu} = \frac{\partial}{\partial \boldsymbol{\theta}} \kappa(\boldsymbol{\theta}) \text{ and } Cov(\mathbf{X}) = \frac{\partial^2 \kappa(\boldsymbol{\theta})}{\partial \boldsymbol{\theta} \partial \boldsymbol{\theta}'} = \left\{ \frac{\partial^2 \kappa(\boldsymbol{\theta})}{\partial \theta_i \partial \theta_j} \right\} = \mathbf{V}.$$

The following proposition provides the ML estimation procedure.

Proposition 1

Consider a random vector $\mathbf{X}$, with probability function belonging to the exponential family. Let $\mathbf{g}(\boldsymbol{\mu})$ be a continuous vector-valued function of $\boldsymbol{\mu}$,

for which the first-order partial derivatives exist. Let $G_\mu = \frac{\partial g(\mu)}{\partial \mu}$ be the derivative of $g(\mu)$ with respect to μ and $G_x = \frac{\partial g(\mu)}{\partial \mu} \big|_{\mu=x}$. The ML estimate of μ subject to the constraints $g(\mu) = 0$, is given by

$$\hat{\mu}_c = x - (G_\mu V)'(G_x V G_\mu')^{-1} g(x) + o(\|x - \mu\|). \quad (2.2)$$

This result implies that the ML estimate must, in general, be obtained iteratively. It is also useful to have the covariance matrix of the ML estimate, $\hat{\mu}_c$.

Proposition 2

The asymptotic covariance matrix of $\hat{\mu}_c$ is given by

$$\Sigma_c = V_\mu - (G_\mu V_\mu)'(G_\mu V_\mu G_\mu')^{-1} G_\mu V_\mu. \quad (2.3)$$

We now show how (2.2) may be used to find ML estimates of the expected frequencies when modelling data in a contingency table in terms of cross-product ratios.

3. The independence model

It has almost become standard practice to model independence in a contingency table in terms of a loglinear model. A more direct approach is to consider the cross-product ratios, also referred to as odds ratios, for the table. Let π_{ij} denote the probability that an observation is classified in cell (i, j) . The set of $(I-1)(J-1)$ local odds ratios for an $I \times J$ contingency table is given by

$$\theta_{ij} = \frac{\pi_{ij}\pi_{i+1,j+1}}{\pi_{i,j+1}\pi_{i+1,j}}, \quad i = 1, \dots, I-1; \quad j = 1, \dots, J-1.$$

The independence model in terms of local odds ratios is

$$\theta_{ij} = 1, \quad \text{for } i = 1, \dots, I-1, \quad j = 1, \dots, J-1.$$

Consider a 3×3 contingency table as an illustration. The model of independence in terms of odds ratios is

$$\begin{aligned} \theta_{11} &= \frac{\pi_{11}\pi_{22}}{\pi_{12}\pi_{21}} = 1, & \theta_{12} &= \frac{\pi_{12}\pi_{23}}{\pi_{13}\pi_{22}} = 1, \\ \theta_{21} &= \frac{\pi_{21}\pi_{32}}{\pi_{22}\pi_{31}} = 1, & \theta_{22} &= \frac{\pi_{22}\pi_{33}}{\pi_{23}\pi_{32}} = 1. \end{aligned}$$

Let F_{ij} denote the expected frequency for cell (i, j) and $\mathbf{F} = (F_{11}, F_{12}, \dots, F_{33})'$ be the vector of expected frequencies. Then the constraints above may be written as

$$\begin{aligned} F_{11}F_{22} - F_{12}F_{21} &= 0, \\ F_{12}F_{23} - F_{13}F_{22} &= 0, \\ F_{21}F_{32} - F_{22}F_{31} &= 0, \\ F_{22}F_{33} - F_{23}F_{32} &= 0. \end{aligned}$$

These constraints are of the form $\mathbf{g}(\mathbf{F}) = \mathbf{0}$. The matrix $\mathbf{G}_F = \frac{\partial \mathbf{g}(\mathbf{F})}{\partial \mathbf{F}}$, where

$$\mathbf{G}_F = \begin{bmatrix} F_{22} & -F_{21} & 0 & -F_{12} & F_{11} & 0 & 0 & 0 & 0 \\ 0 & F_{23} & -F_{22} & 0 & -F_{13} & F_{12} & 0 & 0 & 0 \\ 0 & 0 & 0 & F_{32} & -F_{31} & 0 & -F_{22} & F_{21} & 0 \\ 0 & 0 & 0 & 0 & F_{33} & -F_{32} & 0 & -F_{23} & F_{22} \end{bmatrix}$$

The estimation procedure given in (2.2) may now be applied. If a Poisson sampling procedure is used, then $\mathbf{V} = \mathbf{D}_F$, where $\mathbf{D}_F$ is a diagonal matrix with the elements of $\mathbf{F}$ on the principal diagonal. For multinomial sampling, $\mathbf{V} = \mathbf{D}_F - \frac{1}{n} \mathbf{F} \mathbf{F}'$. Using Poisson sampling, the estimation procedure is

$$\hat{\mathbf{F}}_c = \mathbf{x} - (\mathbf{G}_F \mathbf{D}_F)' (\mathbf{G}_x \mathbf{D}_F \mathbf{G}_F)^{-1} \mathbf{g}(\mathbf{x}) + o(\|\mathbf{x} - \mathbf{F}\|), \quad (3.1)$$

which will require a double iterative procedure, namely over $\mathbf{x}$ and $\mathbf{F}$ to give the ML estimates of the expected frequencies for the independence model. (This double iterative procedure is set out in Matthews and Crowther (1995) and is given in the appendix for an example which follows later.)

We now extend this discussion by considering the loglinear model of conditional independence in a higher order cross-classification. This model is easily expressed in terms of cross-product ratios and the ML estimates for the expected frequencies are found in a way similar to that of independence in a two-way table. As an illustration we shall use a three-way table.

Example 1

Consider the data from Haberman (1978) where the following table was compiled from the 1975 General Social Survey. The table gives the gender of the respondent and level of education in determining attitudes toward roles for women. The question asked is whether "women should take care of running their homes and leave running of the country up to men".

TABLE 1. Subjects cross-classified by attitude towards women staying at home, gender and education of respondent

Sex of respondent	Education of respondent (yrs.)	Response		Total
		Agree	Disagree	
Male	≤ 8	72 (77.4)	47 (41.6)	119
	9 - 12	110 (113.6)	196 (192.4)	306
	≥ 13	44 (36.7)	179 (186.3)	223
		226	422	648
Female	≤ 8	86 (80.6)	38 (43.4)	124
	9 - 12	173 (169.4)	283 (286.6)	456
	≥ 13	28 (35.3)	187 (179.7)	215
		287	508	795
		513	930	$N = 1443$

(Figures in parentheses are the expected frequencies for the conditional independence model.)

- Let S denote the sex of the respondent
 E denote the education level of the respondent
 R denote the response of the respondent.

Now consider the model of conditional independence. Using the notation of Agresti (1984), define the set of local odds ratios as

$$\theta_{ij(k)} = \frac{\pi_{ijk}\pi_{i+1,j+1,k}}{\pi_{i,j+1,k}\pi_{i+1,j,k}}, \quad 1 \leq i \leq r-1, \quad 1 \leq j \leq c-1,$$

for variables X , Y and Z .

The variables X and Y are conditionally independent at level k of Z if all $(r-1)(c-1)$ of the $\theta_{ij(k)}$ at that fixed level of Z are equal to 1. The variables are said to be conditionally independent given Z (i.e. at every level of Z), if all $l(r-1)(c-1)$ of the $\{\theta_{ij(k)}\}$ are equal to 1.

In this example we wish to fit the model of conditional independence of S and R given E . The ML estimates for the expected frequencies in the conditional independence model can be found by using the set of odds ratios. For the above-mentioned conditional independence model

$$\begin{aligned}\theta_{1(1)1} &= \frac{F_{111}F_{212}}{F_{112}F_{211}} = 1, \\ \theta_{1(2)1} &= \frac{F_{121}F_{222}}{F_{122}F_{221}} = 1, \\ \theta_{1(3)1} &= \frac{F_{131}F_{232}}{F_{132}F_{231}} = 1.\end{aligned}$$

The odds ratios $\theta_{1(j)1} = 1$ for $j = 1, 2, 3$ imply

$$\begin{aligned}F_{111}F_{212} - F_{112}F_{211} &= 0, \\ F_{121}F_{222} - F_{122}F_{221} &= 0, \\ F_{131}F_{232} - F_{132}F_{231} &= 0,\end{aligned}$$

which is of the form $\mathbf{g}(\mathbf{F}) = \mathbf{0}$. Once again we find $\mathbf{G}_F = \frac{\partial \mathbf{g}(\mathbf{F})}{\partial \mathbf{F}}$, and using a Poisson sampling procedure, the ML estimates are found by applying

$$\hat{\mathbf{F}}_c = \mathbf{x} - (\mathbf{G}_F \mathbf{D}_F)' (\mathbf{G}_x \mathbf{D}_F \mathbf{G}_F)^{-1} \mathbf{g}(\mathbf{x}) + o(\|\mathbf{x} - \mathbf{F}\|),$$

which will require a double iterative procedure, namely over $\mathbf{x}$ and $\mathbf{F}$.

For the conditional independence model, the likelihood ratio statistic, $LR = 6.02$ and the Pearson chi-squared statistic, $X^2 = 5.99$ based on $df = 3$ with attained significance level, $p = 0.11$, showing that the fit is reasonable. Note that df = number of constraints in the function $\mathbf{g}(\mathbf{F})$.

An advantage of the proposed estimation procedure is that zero or small frequencies do not present a problem, since the logarithms are not required as we are not using the loglinear form of the model in the estimation algorithm.

4. Quasi-independence in the $I \times J$ table

We now show how the constraints implied by the quasi-independence (QI) model may be used to find the ML estimates of the expected frequencies. Using the notation of Goodman (1994), let the odds-ratio $\theta_{ij,rs}$ be defined as follows:

$$\theta_{ij,rs} = \frac{\pi_{ij}\pi_{rs}}{\pi_{is}\pi_{rj}} \text{ for } i < r \text{ and } j < s. \quad (4.1)$$

Instead of considering all possible odds ratios for the $I \times J$ table, consider the situation where a given subset S of the cells (i, j) in the table, is of interest. The concept of quasi-independence is stated as

$$\pi_{ij} = \alpha_i \beta_j, \quad \forall \text{ cells } (i, j) \text{ in } S. \quad (4.2)$$

From equation (4.1) the QI model implies that

$$\theta_{ij,rs} = \frac{\alpha_i \beta_j \alpha_r \beta_s}{\alpha_i \beta_s \alpha_r \beta_j} = 1, \quad \forall T_{ij,rs} \text{ in } T \quad (4.3)$$

where T is the set of 2×2 tables $T_{ij,rs}$ in which cells (i, j) , (r, s) , (i, s) and (r, j) are in S .

Once again the conditions given in (4.3) can be written in terms of constraints $\mathbf{g}(\boldsymbol{\pi}) = \mathbf{0}$ and the ML estimation procedure of proposition 1 can be employed to find the ML estimates of the expected frequencies in the QI model.

Example 2

Consider the migration example of Agresti (1990) where the table compares region of residence in 1985 with 1980 for a sample of US residents. Consider whether, for people who moved, residence in 1985 is independent of region in 1980. We investigate independence for cells not on the principal diagonal, that is a subtable S in T .

TABLE II. Migration from 1980 to 1985

Residence in 1980	Residence in 1985			
	North East	Mid West	South	West
North East	11607	100 (126.6)	366 (312.9)	124 (150.5)
Mid West	87 (117.4)	13677	515 (531.1)	302 (255.5)
South	172 (133.2)	225 (243.8)	17819	270 (290.0)
West	63 (71.4)	176 (130.6)	286 (323.0)	10192

(Figures in parentheses are the ML estimates of the expected frequencies for the QI model.)

Let $\mathbf{F} = (F_1, F_2, \dots, F_{16})'$ denote the expected frequencies arranged as a column vector, where $F_1 = F_{11}, F_2 = F_{12}, \dots, F_{16} = F_{44}$. For the quasi-independence model there are 5 cross-product ratios, which are each set equal to one. These are given by

$$\frac{F_5 F_{12}}{F_8 F_9} = 1, \frac{F_3 F_8}{F_4 F_7} = 1, \frac{F_2 F_{15}}{F_3 F_{14}} = 1, \frac{F_5 F_{15}}{F_7 F_{13}} = 1 \text{ and } \frac{F_9 F_{14}}{F_{10} F_{13}} = 1.$$

These constraints can now be expressed in the form $\mathbf{g}(\mathbf{F}) = \mathbf{0}$, where

$$g_1(\mathbf{F}) = F_5 F_{12} - F_8 F_9 = 0,$$

$$g_2(\mathbf{F}) = F_3 F_8 - F_4 F_7 = 0,$$

$$g_3(\mathbf{F}) = F_2 F_{15} - F_3 F_{14} = 0,$$

$$g_4(\mathbf{F}) = F_5 F_{15} - F_7 F_{13} = 0$$

$$\text{and } g_5(\mathbf{F}) = F_9 F_{14} - F_{10} F_{13} = 0.$$

The matrix $\mathbf{G}_F = \frac{\partial \mathbf{g}(\mathbf{F})}{\partial \mathbf{F}}$ has 5 rows and 16 columns. Let $\mathbf{G}_F(i, j)$ denote the element in the i th row and j th column. Then

$$\begin{array}{llll} \mathbf{G}_F(1, 5) = F_{12}, & \mathbf{G}_F(1, 8) = -F_9, & \mathbf{G}_F(1, 9) = -F_8, & \mathbf{G}_F(1, 12) = F_5, \\ \mathbf{G}_F(2, 3) = F_8, & \mathbf{G}_F(2, 4) = -F_7, & \mathbf{G}_F(2, 7) = -F_4, & \mathbf{G}_F(2, 8) = F_3, \\ \mathbf{G}_F(3, 2) = F_{15}, & \mathbf{G}_F(3, 3) = -F_{14}, & \mathbf{G}_F(3, 14) = -F_3, & \mathbf{G}_F(3, 15) = F_2, \\ \mathbf{G}_F(4, 5) = F_{15}, & \mathbf{G}_F(4, 7) = -F_{13}, & \mathbf{G}_F(4, 13) = -F_7, & \mathbf{G}_F(4, 15) = F_5, \\ \mathbf{G}_F(5, 9) = F_{14}, & \mathbf{G}_F(5, 10) = -F_{13}, & \mathbf{G}_F(5, 13) = -F_{10}, & \mathbf{G}_F(5, 14) = F_9. \end{array}$$

All other $\mathbf{G}_F(i, j)$ are zero. The ML estimation procedure

$$\hat{\mathbf{F}}_c = \mathbf{x} - (\mathbf{G}_F \mathbf{D}_F)' (\mathbf{G}_x \mathbf{D}_F \mathbf{G}_F')^{-1} \mathbf{g}(\mathbf{x}) + o(\|\mathbf{x} - \mathbf{F}\|)$$

is implemented and the double iterative procedure is required for convergence to the ML estimates of the expected frequencies for the QI model. The likelihood ratio, $LR = 69.11$ with $df = 5$.

5. Quasi-independence in an $R \times R$ triangular table

Goodman (1994) considers the problem of quasi-independence in an $R \times R$ triangular table. In the $R \times R$ triangular table, let π_{ij} denote the probability that an observation will fall in the i th row and j th column of the table. The QI model states that

$$\begin{aligned} \pi_{ij} &= \alpha_i \beta_j, & \text{for } i \leq j, \\ &= 0, & \text{for } i > j, \end{aligned}$$

with $\alpha_i > 0$ and $\beta_j > 0$, for $i = 1, \dots, R$, and $j = 1, \dots, R$. Without loss of generality consider the table with positive probabilities in the upper-right triangle. Goodman presents an explicit expression in terms of the row and

column marginal proportions, which gives the ML estimates of the frequencies expected under the quasi-independence model for the triangular table.

The problem of QI can also be easily modelled in terms of all possible odds ratios for the triangular table set equal to one, which can be expressed in terms of constraints on the cell probabilities. We use the following example to illustrate the procedure.

Example 3

Consider the 5×5 contingency table taken from Bishop, Fienberg and Holland (1975) wherein the initial and final condition of stroke patients is considered.

TABLE III. Initial and final condition of stroke patients

Initial condition	Final condition				
	A	B	C	D	E
E	11 (15.66)	23 (21.92)	12 (11.93)	15 (11.48)	8 (8.0)
D	9 (6.16)	10 (8.63)	4 (4.69)	1 (4.52)	0 (0)
C	6 (4.43)	4 (6.20)	4 (3.37)	0 (0)	0 (0)
B	4 (3.75)	5 (5.25)	0 (0)	0 (0)	0 (0)
A	5 (5.0)	0 (0)	0 (0)	0 (0)	0 (0)

(Figures in parentheses are the ML estimates of the expected frequencies for the QI model.)

For convenience arrange the cell probabilities π_{ij} as a vector $\pi = (\pi_1, \pi_2, \dots, \pi_{25})'$, where $\pi_1 = \pi_{11}$, $\pi_2 = \pi_{12}$, ..., $\pi_{25} = \pi_{55}$.

The QI model can be expressed in terms of 6 cross-product ratios, each of which must be equal to one, for independence to hold for the non-zero cells.

The constraints are

$$\frac{\pi_1\pi_7}{\pi_2\pi_6} = 1, \quad \frac{\pi_1\pi_8}{\pi_3\pi_6} = 1, \quad \frac{\pi_1\pi_9}{\pi_4\pi_6} = 1, \quad \frac{\pi_1\pi_{12}}{\pi_2\pi_{11}} = 1, \quad \frac{\pi_1\pi_{13}}{\pi_3\pi_{11}} = 1, \quad \frac{\pi_1\pi_{17}}{\pi_2\pi_{16}} = 1.$$

These constraints in turn can now be expressed in the form $g(\pi) = 0$, where

$$\begin{aligned} g_1(\pi) &= \pi_1\pi_7 - \pi_2\pi_6 = 0, \\ g_2(\pi) &= \pi_1\pi_8 - \pi_3\pi_6 = 0, \\ g_3(\pi) &= \pi_1\pi_9 - \pi_4\pi_6 = 0, \\ g_4(\pi) &= \pi_1\pi_{12} - \pi_2\pi_{11} = 0, \\ g_5(\pi) &= \pi_1\pi_{13} - \pi_3\pi_{11} = 0, \\ g_6(\pi) &= \pi_1\pi_{17} - \pi_2\pi_{16} = 0. \end{aligned}$$

The matrix $G_\pi = \frac{\partial g(\pi)}{\partial \pi}$ is a 6×25 matrix, with

$$\begin{aligned} G_\pi(1, 1) &= \pi_7, & G_\pi(1, 2) &= -\pi_6, & G_\pi(1, 6) &= -\pi_2, & G_\pi(1, 7) &= \pi_1, \\ G_\pi(2, 1) &= \pi_8, & G_\pi(2, 3) &= -\pi_6, & G_\pi(2, 6) &= -\pi_3, & G_\pi(2, 8) &= \pi_1, \\ G_\pi(3, 1) &= \pi_9, & G_\pi(3, 4) &= -\pi_6, & G_\pi(3, 6) &= -\pi_4, & G_\pi(3, 9) &= \pi_1, \\ G_\pi(4, 1) &= \pi_{12}, & G_\pi(4, 2) &= -\pi_{11}, & G_\pi(4, 11) &= -\pi_2, & G_\pi(4, 12) &= \pi_1, \\ G_\pi(5, 1) &= \pi_{13}, & G_\pi(5, 3) &= -\pi_{11}, & G_\pi(5, 11) &= -\pi_3, & G_\pi(5, 13) &= \pi_1, \\ G_\pi(6, 1) &= \pi_{17}, & G_\pi(6, 2) &= -\pi_{16}, & G_\pi(6, 16) &= -\pi_2, & G_\pi(6, 17) &= \pi_1, \end{aligned}$$

All other entries $G_\pi(i, j)$ are equal to zero.

Let $F = (F_1, \dots, F_{25})'$ denote the vector of expected frequencies and n be the total number of observations, then for example

$$\frac{\pi_1\pi_7}{\pi_2\pi_6} \text{ may be written as } \frac{(n\pi_1)(n\pi_7)}{(n\pi_2)(n\pi_6)} = 1 \text{ or } \frac{F_1F_7}{F_2F_6} = 1$$

and similarly for the other cross-product ratios. The constraints $\mathbf{g}(\boldsymbol{\pi}) = 0$ may be written in the form $\mathbf{g}(\mathbf{F}) = 0$ and $\mathbf{G}_F = \frac{\partial \mathbf{g}(\mathbf{F})}{\partial \mathbf{F}}$ is the 6×25 matrix with elements of $\mathbf{G}_\pi$ replaced by the corresponding entry in terms of the expected frequencies, that is $\mathbf{G}_F(1, 1) = F_7$ and so on.

The ML estimation procedure

$$\hat{\mathbf{F}}_c = \mathbf{x} - (\mathbf{G}_F \mathbf{D}_F)' (\mathbf{G}_x \mathbf{D}_F \mathbf{G}_F')^{-1} \mathbf{g}(\mathbf{x}) + o(\|\mathbf{x} - \mathbf{F}\|)$$

may now be used to find the ML estimates of the expected frequencies in the QI model. Once again the double iterative procedure will provide the ML estimates on convergence. The likelihood ratio statistic, $LR = 9.60$ with $df = 6$.

6. Quasi-symmetry

The quasi-symmetry model has multiplicative form

$$\pi_{ij} = \alpha_i \beta_j \gamma_{ij}, \text{ where } \gamma_{ij} = \gamma_{ji} \quad \forall i, j, \text{ and } \alpha_i, \beta_j, \gamma_{ij} > 0.$$

Quasi-symmetry can also be written in terms of symmetry of odds ratios. The quasi-symmetry model holds if and only if for all integers i, j, k and l between 1 and R

$$\frac{\pi_{ik}\pi_{jl}}{\pi_{il}\pi_{jk}} = \frac{\pi_{ki}\pi_{lj}}{\pi_{li}\pi_{kj}}. \quad (6.1)$$

Now fixing one cell, by taking $j = l = R$ and forming all odds ratios, equation (6.1) becomes

$$\frac{\pi_{ik}\pi_{RR}}{\pi_{iR}\pi_{Rk}} = \frac{\pi_{ki}\pi_{RR}}{\pi_{Ri}\pi_{kR}} \quad \text{or} \quad \pi_{ik}\pi_{kR}\pi_{Ri} = \pi_{ki}\pi_{iR}\pi_{Rk}.$$

These constraints can now be written in the form $g(\pi) = 0$ and the estimation procedure may be used to find the ML estimates of the expected frequencies for the quasi-symmetry model.

Example 4

Consider the 4×4 migration table given in table II. For the quasi-symmetry model to hold, the following constraints must hold:

$$\begin{array}{ll} \pi_{12}\pi_{24}\pi_{41} = \pi_{21}\pi_{14}\pi_{42} & F_{12}F_{24}F_{41} - F_{21}F_{14}F_{42} = 0 \\ \pi_{13}\pi_{34}\pi_{41} = \pi_{31}\pi_{14}\pi_{43} & \text{or} \quad F_{13}F_{34}F_{41} - F_{31}F_{14}F_{43} = 0 \\ \pi_{23}\pi_{34}\pi_{42} = \pi_{32}\pi_{24}\pi_{43} & F_{23}F_{34}F_{42} - F_{32}F_{24}F_{43} = 0 \end{array}$$

which is of the form $g(F) = 0$ and $G_F = \frac{\partial g(F)}{\partial F}$ is a 3×16 matrix, with

$$\begin{array}{lll} G_F(1, 2) = F_{24}F_{41}, & G_F(1, 4) = -F_{21}F_{42}, & G_F(1, 5) = -F_{14}F_{42}, \\ G_F(1, 8) = F_{12}F_{41}, & G_F(1, 13) = F_{12}F_{24}, & G_F(1, 14) = -F_{21}F_{14}, \\ \\ G_F(2, 3) = F_{34}F_{41}, & G_F(2, 4) = -F_{31}F_{43}, & G_F(2, 9) = -F_{14}F_{43}, \\ G_F(2, 12) = F_{13}F_{41}, & G_F(2, 13) = F_{13}F_{34}, & G_F(2, 15) = -F_{31}F_{14}, \\ \\ G_F(3, 7) = F_{34}F_{42}, & G_F(3, 8) = -F_{32}F_{43}, & G_F(3, 10) = -F_{24}F_{43}, \\ G_F(3, 12) = F_{23}F_{42}, & G_F(3, 14) = F_{23}F_{34}, & G_F(3, 15) = -F_{32}F_{24}. \end{array}$$

All other $G_F(i, j)$ are zero.

The ML estimates of the expected frequencies are found by applying

$$\hat{\mathbf{F}}_c = \mathbf{x} - (\mathbf{G}_F \mathbf{D}_F)' (\mathbf{G}_x \mathbf{D}_F \mathbf{G}_F')^{-1} \mathbf{g}(\mathbf{x}) + o(\|\mathbf{x} - \mathbf{F}\|).$$

A double iteration over $\mathbf{x}$ and $\mathbf{F}$ is necessary. The ML estimates for the quasi-symmetry model are given in parentheses in the following table.

TABLE IV Migration from 1980 to 1985

Residence in 1980	Residence in 1985			
	North East	Mid West	South	West
North East	11607	100 (95.8)	366 (370.4)	124 (123.8)
Mid West	87 (91.2)	13677	515 (501.7)	302 (311.1)
South	172 (167.6)	225 (238.3)	17819	270 (261.1)
West	63 (63.2)	176 (166.9)	286 (294.9)	10192

The likelihood ratio statistic, $LR = 2.99$ with $df = 3$ and attained significance level, $p = 0.39$, showing that the model fits the data reasonably well.

7. Mixed models

A distinct advantage of the estimation procedure of proposition 1 over the standard estimation procedures used in many statistical packages is the ability to deal with cases where the constraints on the frequencies may not be expressed as a parametric model of the form say, $h(\mathbf{F}) = \mathbf{X}\beta$, or tables for which different sets of restrictions hold over regions of the table. Consider for example quasi-independence for a 4×4 table along with two symmetry constraints, namely $F_{21} = F_{12}$ and $F_{43} = F_{34}$. The quasi-independence constraints are those previously stated in example 2. These constraints along with the

symmetry constraints can be written in the form $\mathbf{g}(\mathbf{F}) = \mathbf{0}$, with elements $g_1(\mathbf{F}) = 0, \dots, g_5(\mathbf{F}) = 0$, from example 2 and

$$\begin{aligned}g_6(\mathbf{F}) &= F_5 - F_2 = 0, \\g_7(\mathbf{F}) &= F_{15} - F_{12} = 0,\end{aligned}$$

where we continue with the subscripting adopted in example 2, namely, $\mathbf{F} = (F_1, F_2, \dots, F_{16})'$, where $F_1 = F_{11}, F_2 = F_{12}, \dots, F_{16} = F_{44}$.

The matrix $\mathbf{G}_F = \frac{\partial \mathbf{g}(\mathbf{F})}{\partial \mathbf{F}}$ has 7 rows and 16 columns. Let $\mathbf{G}_F(i, j)$ denote the element in the i th row and j th column. Then the elements in the first five rows are those of $\mathbf{G}_F$ in example 2 and the elements of rows 6 and 7 are

$$\begin{aligned}\mathbf{G}_F(6, 2) &= -1, & \mathbf{G}_F(6, 5) &= 1, \\ \mathbf{G}_F(7, 12) &= -1, & \mathbf{G}_F(7, 15) &= 1.\end{aligned}$$

All other $\mathbf{G}_F(i, j)$ are zero. The ML estimation procedure

$$\hat{\mathbf{F}}_c = \mathbf{x} - (\mathbf{G}_F \mathbf{D}_F)' (\mathbf{G}_x \mathbf{D}_F \mathbf{G}_F')^{-1} \mathbf{g}(\mathbf{x}) + o(\|\mathbf{x} - \mathbf{F}\|)$$

is implemented and the double iterative procedure is required for convergence to the ML estimates.

Example 5

Consider the frequency table in example 2 and apply the above-mentioned quasi-independence and symmetry constraints to the table. The two symmetry constraints imply that the same number of people move from Mid West to North East as North East to Mid West and the same number move from West to South as South to West.

TABLE V Migration from 1980 to 1985

Residence in 1980	Residence in 1985			
	North East	Mid West	South	West
North East	11607	100 (121.8) ^a (130.5) ^b	366 (304.9) (218.3)	124 (151.9) (107.2)
Mid West	87 (121.8) (130.5)	13677	515 (529.8) (383.7)	302 (263.9) (188.3)
South	172 (141.3) (218.3)	225 (245.5) (383.7)	17819	270 (306.3) (315.0)
West	63 (70.4) (107.2)	176 (122.3) (188.3)	286 (306.3) (315.0)	10192

(Figures in parentheses are the ML estimates of the expected frequencies for the mixed models, ^a quasi-independence with two symmetry constraints and ^b quasi-independence with three symmetry constraints.)

The Pearson chi-squared statistic for the quasi-independence model with two symmetry constraints is $X^2 = 75.5$ and the likelihood ratio statistic, $LR = 73.2$ with $df = 7$. If we impose a further symmetry constraint, say $F_{32} = F_{23}$ on the table, then it is interesting to note that the constraints impose a form of symmetry on the expected frequencies for the whole table. This, however, is not the usual model of symmetry, for which the expected frequency for the cells (i, j) and (j, i) , $i \neq j$, is simply the average of the observed frequencies for these two cells. The Pearson chi-squared statistic for the quasi-independence model with three symmetry constraints is $X^2 = 341.3$ and the likelihood ratio statistic, $LR = 327.6$ with $df = 8$. Although the fit is not good, the constraints produce an interesting result.

This example illustrates the flexibility of the estimation procedure and ML estimation for this type of problem is not catered for in standard statistical packages.

A program implementing the estimation procedure for quasi-independence with two symmetry constraints appears in the appendix. The program makes use of SAS/IML, which is a procedure of the statistical package SAS. The SAS/IML procedure is a powerful, flexible interactive matrix language. The programs for the other examples follow the same algorithm.

8. Conclusion

The estimation procedure of proposition 1 proves useful for finding the ML estimates for the expected frequencies when modelling in terms of the cross-product ratios. It is easy to express the constraints on the cross-product ratios as some function $g(\mathbf{F}) = 0$, which satisfies the conditions of the estimation procedure of proposition 1. The estimation procedure also caters for non-standard models (such as the mixed model previously discussed), which are not a feature of standard statistical packages.

9. References

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10. Appendix

The following SAS/IML program is for example 5.

```
proc iml worksizes=200;
reset nolog;
x={ 11607,100,366,124,
    87,13677,515,302,
    172,225,17818,270,
    63,176,286,10192};
n=x[+];
x11=x;
Gx=J(7,16,0);
Gm=J(7,16,0);
x1=x;
diff1=1;
i=0;
do while (diff1>0.000001);
i=i+1;
m=x;
x=x1;
V=diag(m)-m*m'/n;
Gm[1,5]= m[12]; Gm[1,8]=-m[9]; Gm[1,9]=-m[8];
Gm[1,12]= m[5];
Gm[2,3]= m[8]; Gm[2,4]=-m[7]; Gm[2,7]=-m[4];
Gm[2,8]= m[3];
Gm[3,2]= m[15]; Gm[3,3]=-m[14]; Gm[3,14]=-m[3];
Gm[3,15]= m[2];
Gm[4,5]= m[15]; Gm[4,7]=-m[13]; Gm[4,13]=-m[7];
```

```

Gm[4,15] = m[5];
Gm[5,9] = m[14]; Gm[5,10] = -m[13]; Gm[5,13] = -m[10];
Gm[5,14] = m[9];
Gm[6,2] = -1; Gm[6,5] = 1;
Gm[7,12] = -1; Gm[7,15] = 1;
    diff=1;
j=0;
do while (diff>0.000001);
    xd=x;
j=j+1;
g1=x[5]*x[12]-x[8]*x[9];
g2=x[3]*x[8]-x[4]*x[7];
g3=x[2]*x[15]-x[3]*x[14];
g4=x[5]*x[15]-x[7]*x[13];
g5=x[9]*x[14]-x[10]*x[13];
g6=x[5]-x[2];
g7=x[15]-x[12];
g=g1//g2//g3//g4//g5//g6//g7;
Gx[1,5] = x[12]; Gx[1,8] = -x[9]; Gx[1,9] = -x[8];
Gx[1,12] = x[5];
Gx[2,3] = x[8]; Gx[2,4] = -x[7]; Gx[2,7] = -x[4];
Gx[2,8] = x[3];
Gx[3,2] = x[15]; Gx[3,3] = -x[14]; Gx[3,14] = -x[3];
Gx[3,15] = x[2];
Gx[4,5] = x[15]; Gx[4,7] = -x[13]; Gx[4,13] = -x[7];
Gx[4,15] = x[5];
Gx[5,9] = x[14]; Gx[5,10] = -x[13]; Gx[5,13] = -x[10];
Gx[5,14] = x[9];
Gx[6,2] = -1; Gx[6,5] = 1;
Gx[7,12] = -1; Gx[7,15] = 1;
x=x-(Gm*V)'*ginv(Gx*V*Gm')*g;
diff=(x-xd)'*(x-xd);

```

```
nt=x[+];  
end;  
diff1=(m-x)'*(m-x);  
end;  
print i j x x11;  
df=nrow(Gx) ;  
ch2=(x11-m)'*((x11-m)/m);  
G2=2*x11'*log(x11/m);  
vec=ch2||G2;  
prob= J(1,2,1)-probchi(vec,df);  
print vec prob;
```